

Design FMEA (qualitative version) — VBF-T4R1-DFMEA-01

Qualitative version, based on simulation signals; severity/occurrence rated high/medium/low from the magnitude of the simulated value vs threshold. RPN = qualitative S×O matrix.

Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
Negative electrode lithium plating (fast charge)	Electrolyte salt depletion → local conductivity/diffusivity collapse at anode end-of-charge	anode_potential_v min: baseline −184.8 mV → final design +20.5 mV	high (dendrite → internal short)	low (positive margin in final design; was high in rounds 3–5)	LHCE transport profile (κ 1.1 flat, D_e 3e-10 flat, t 0.6), fine graphite (2.5 μ m) exchange area, h = 80 keeps 45 °C charge cool enough
Thermal runaway risk (temperature rise exceeds limit)	Polarization heat at 4C/45 °C charge	T_max_K 327.70 K vs 333.15 K limit (margin 5.4 K)	high (thermal runaway)	low	Immersion cooling h = 80 W/m ² /K; lower polarization heat via transport fixes
Electrolyte oxidative decomposition (voltage window)	Electrolyte HOMO vs cathode potential	Not quantified — real_compute=false, no DFT HOMO/IE-EA endorsement ran	medium	low	Standard window 2.5–4.2 V below typical EC-class oxidation onset (domain experience); recommend DFT endorsement before production
Insufficient capacity	Active material / geometry shortfall	1C capacity 5.0368 Ah vs nominal 5.0 Ah (100.7 %)	medium	low	Verified at 1C; 4C charge voltage-limited (4.2 V after 0.419 Ah runner-scale) — acceptable per criteria
−20 °C power/capacity loss (extreme-cold mission)	Electrolyte transport freeze-out	retention 0.99267 (cold-soak, no self-heating) — but computed under flat-transport idealization	medium	medium (model optimism: real LHCE κ (−20 °C) $\approx$ 0.3 S/m estimate; margin 4.27 pt)	Confirm electrolyte −20 °C conductivity by measurement; keep self-heating margin if h reduced

Conclusion

Highest-risk items: −20 °C electrolyte transport optimism (occurrence medium — the only item without a comfortable simulation margin once real transport properties are considered) and electrolyte oxidation stability (not quantified without true-compute endorsement). Both have mitigations recorded; the plating and thermal items that dominated the design effort (rounds 3–6) are mitigated in the final design with positive margins. Complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary).